

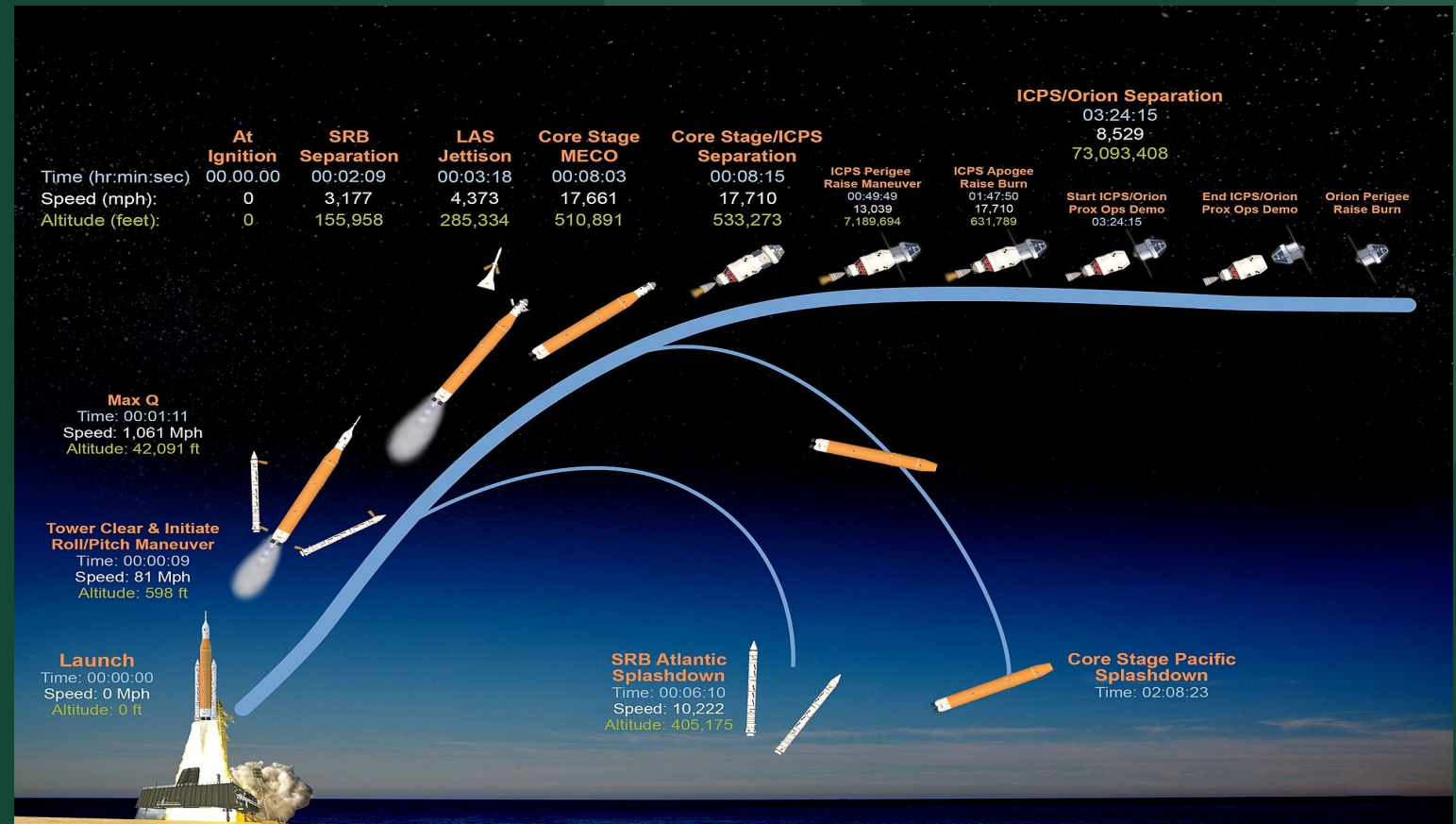
Artemis II Rocket Nozzle Performance During Mission

Alex Rubin, Miles Feheley, Loa Georgsdottir

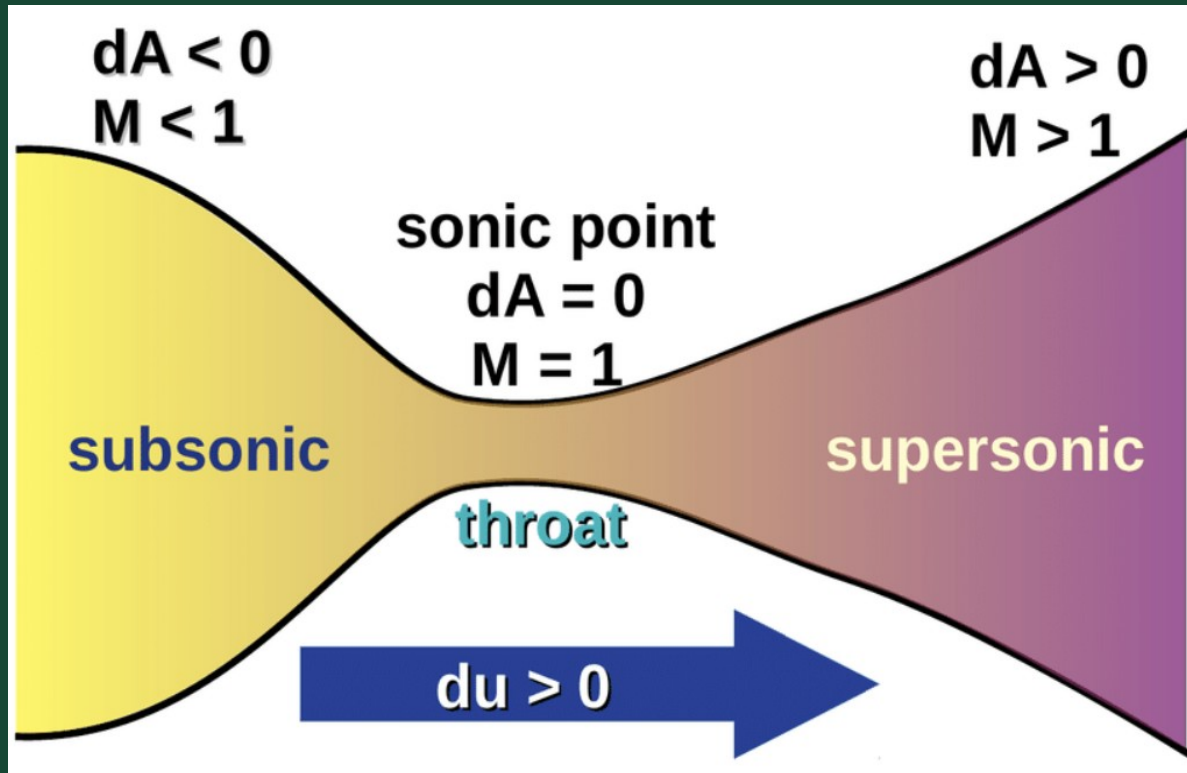
Motivation + Background

Studying fixed-area rocket nozzle performance during core stage ascent

- Realistic flight
- Optimization
- Engine test + Validation



Physical Model



Flow:

- $M = 1$ at throat
- Expansion into supersonic regime in diverging section
- Flow driven by pressure ratio
 - o $P_{\text{chamber}} / P_{\text{ambient}}$

Key Parameters:

- Mach Number
- Reynolds number
- Pressure ratio

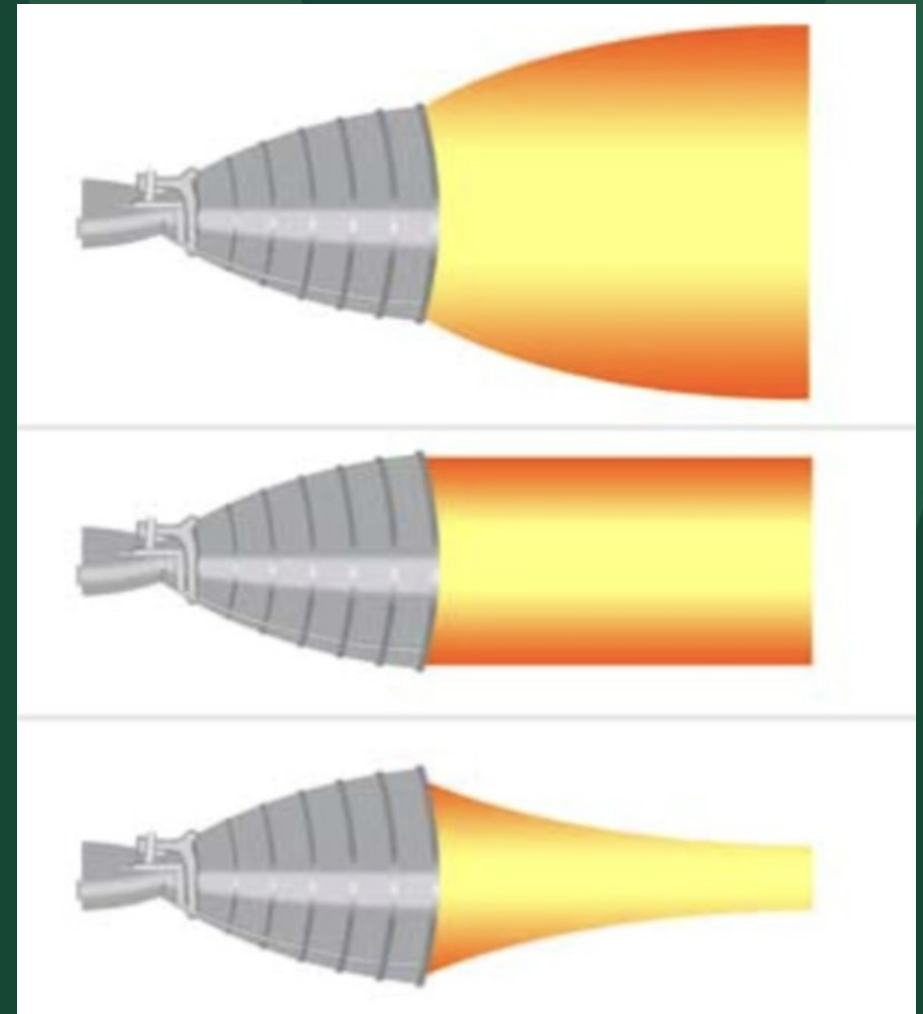
Physical Model

Expansion regimes:

- under expanded $P_{\text{exit}} > P_{\text{ambient}}$
- Ideally expanded: $P_{\text{exit}} = P_{\text{ambient}}$
- Overexpanded $P_{\text{exit}} < P_{\text{ambient}}$

Efficiency:

- Thrust max when $P_{\text{exit}} = P_{\text{ambient}}$
- Losses: shocks, separation, thrust loss
- $\uparrow$ altitude $\rightarrow \downarrow P_{\text{ambient}} \rightarrow$ improved expansion



Assumptions

Physical

- No heat transfer along the walls of the nozzle or cooling systems (Adiabatic)
- No regeneration for preheating of the fuel and oxidizer
- Ideal Gas law is used for the exhaust fluid
- Rocket's vertical velocity and acceleration are fixed (defined by UDF trajectory)

Numerical

- 2D Axisymmetric (Simplified from 3D for computational efficiency)
- Ambient pressure varies with altitude via User-Defined Function (UDF)

Numerical Model + Methods

- Density-based, Axisymmetric, Transient Solver
- K-omega SST Turbulence model
- Formulation: Implicit
- Flux Type: Roe-FDS
- Discretization
 - Gradient = Least squares cell based
 - Flow = First order upwind
 - Pseudo time method = off
- Number of Iterations = Variable

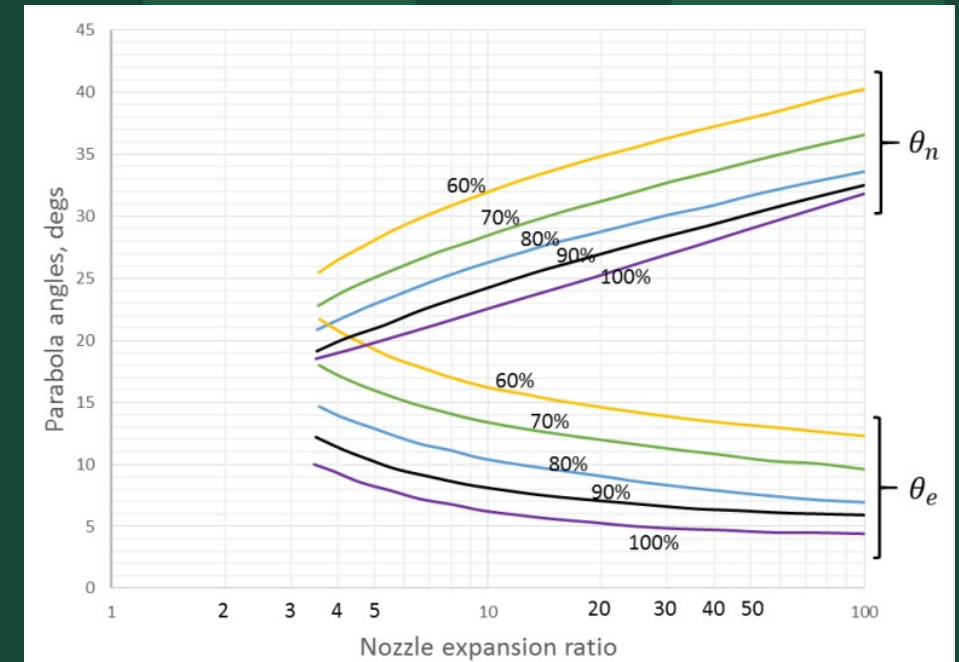
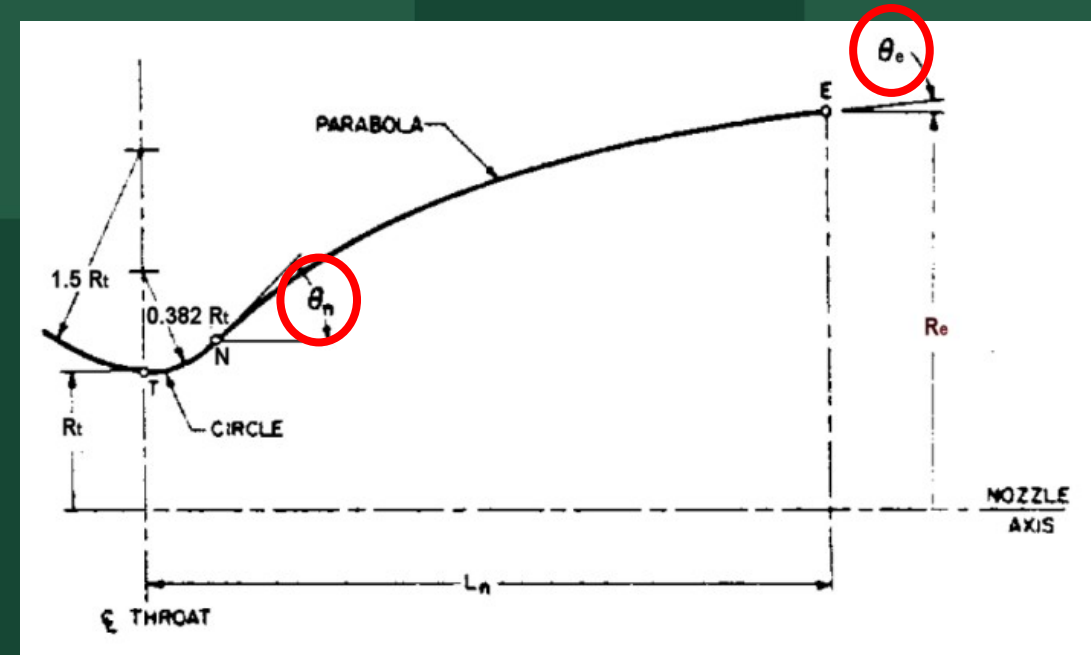
Nozzle Design

RS-25 Nozzle – Not a published geometry

Nozzle geometries were created using known RS Nozzle Shaping Functions plus known quantities:

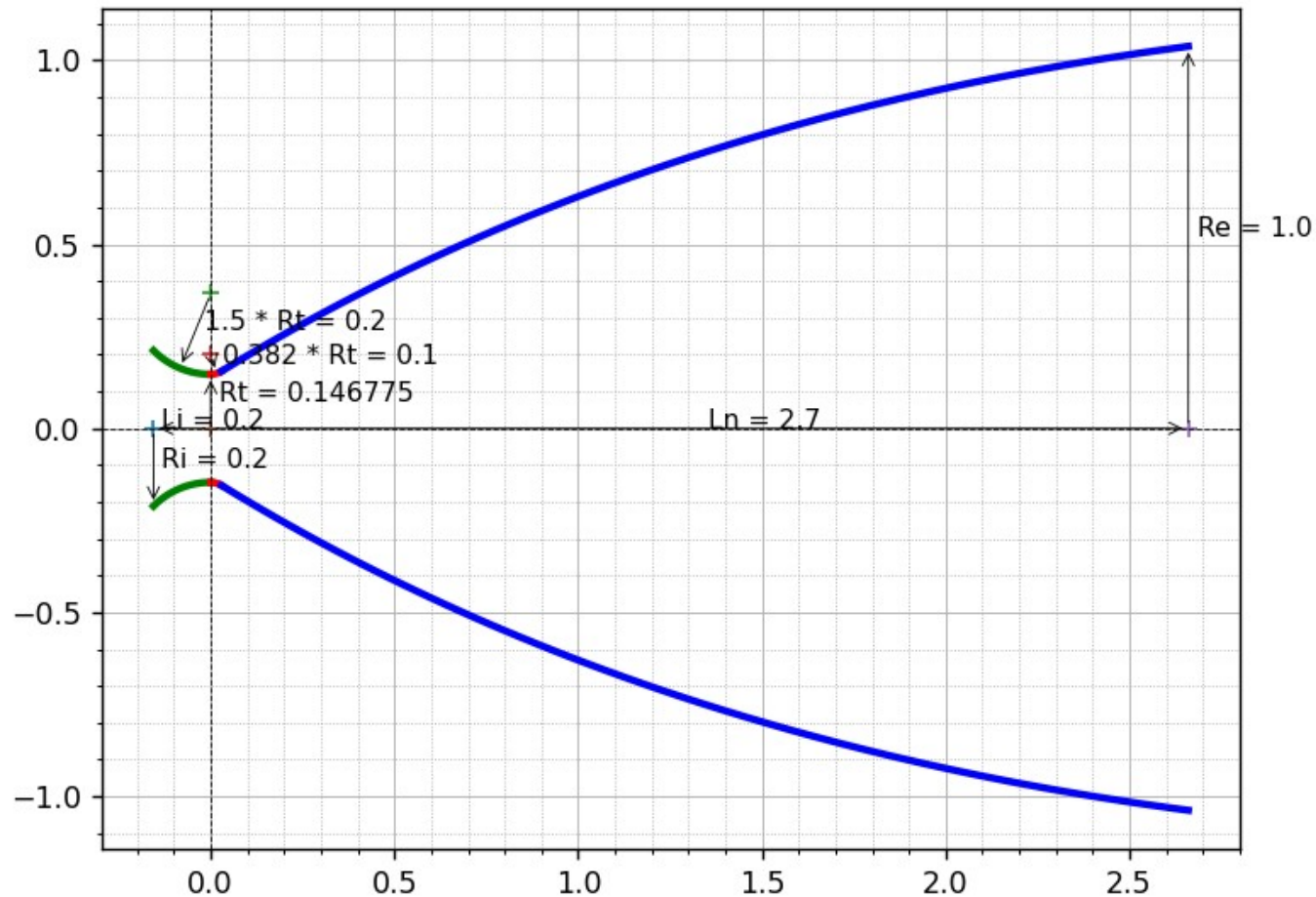
$$D_{\text{throat}} = 0.29355\text{m}$$

$$L_{\text{throat to exit}} = 4.2672\text{m}$$

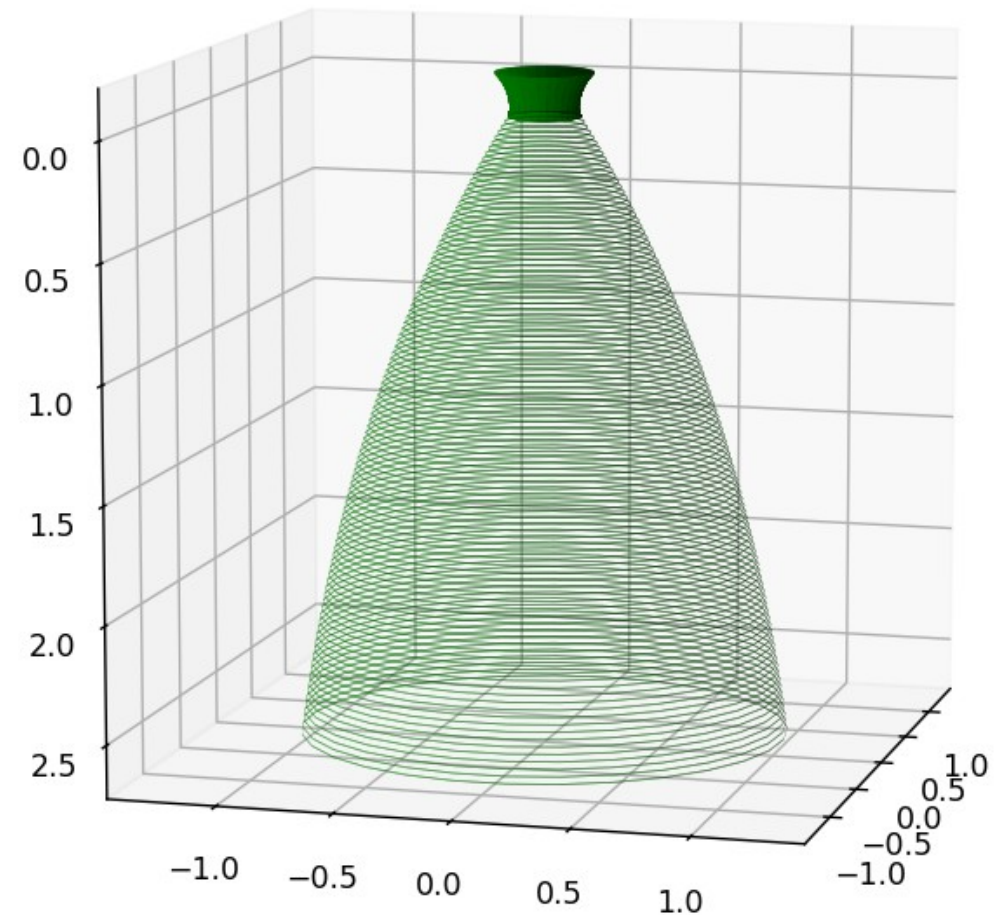


Nozzle Design: Area Ratio = 50

Bell Nozzle
[Area Ratio = 50, Throat Radius = 0.146775]

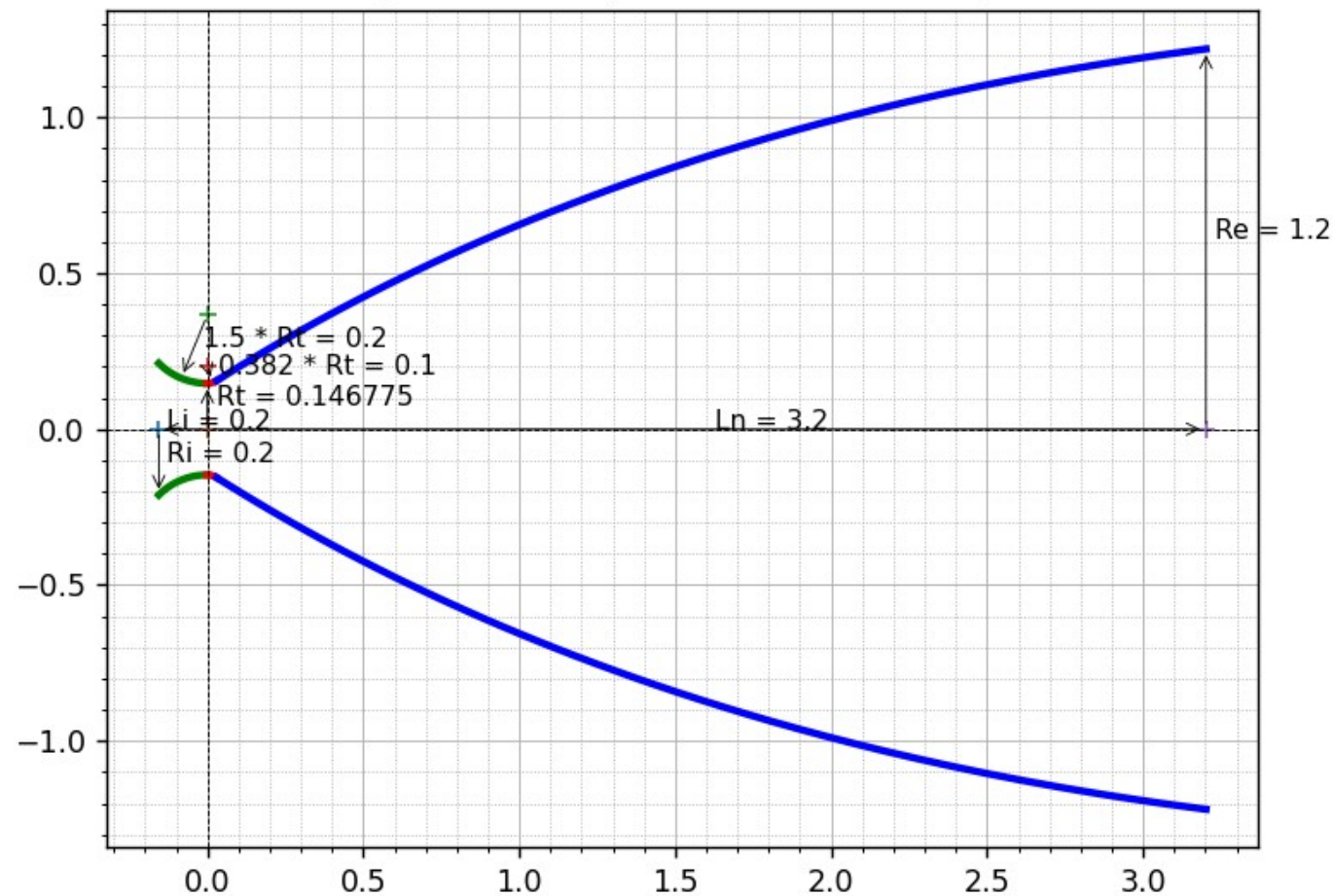


$$\theta_E = 0.1308996938995747, \theta_N = 0.5497787143782138$$

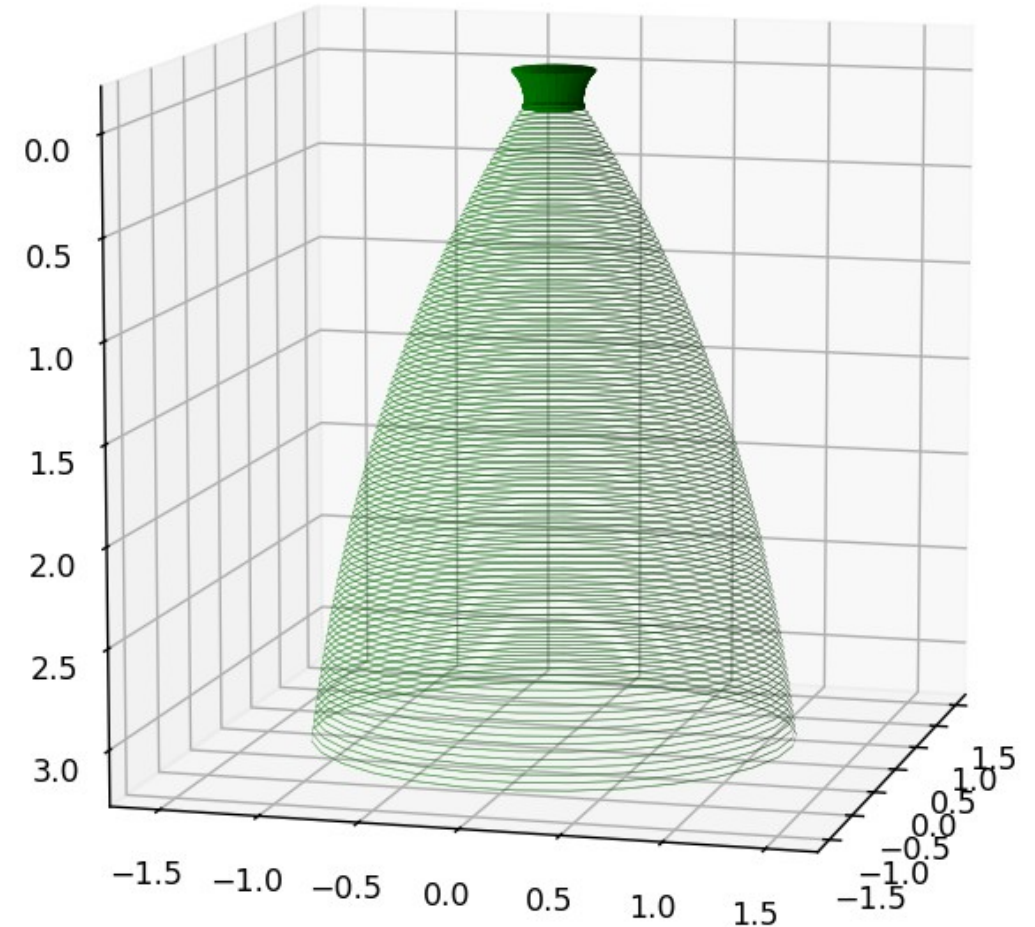


Nozzle Design: Area Ratio = 69

Bell Nozzle
[Area Ratio = 69, Throat Radius = 0.146775]

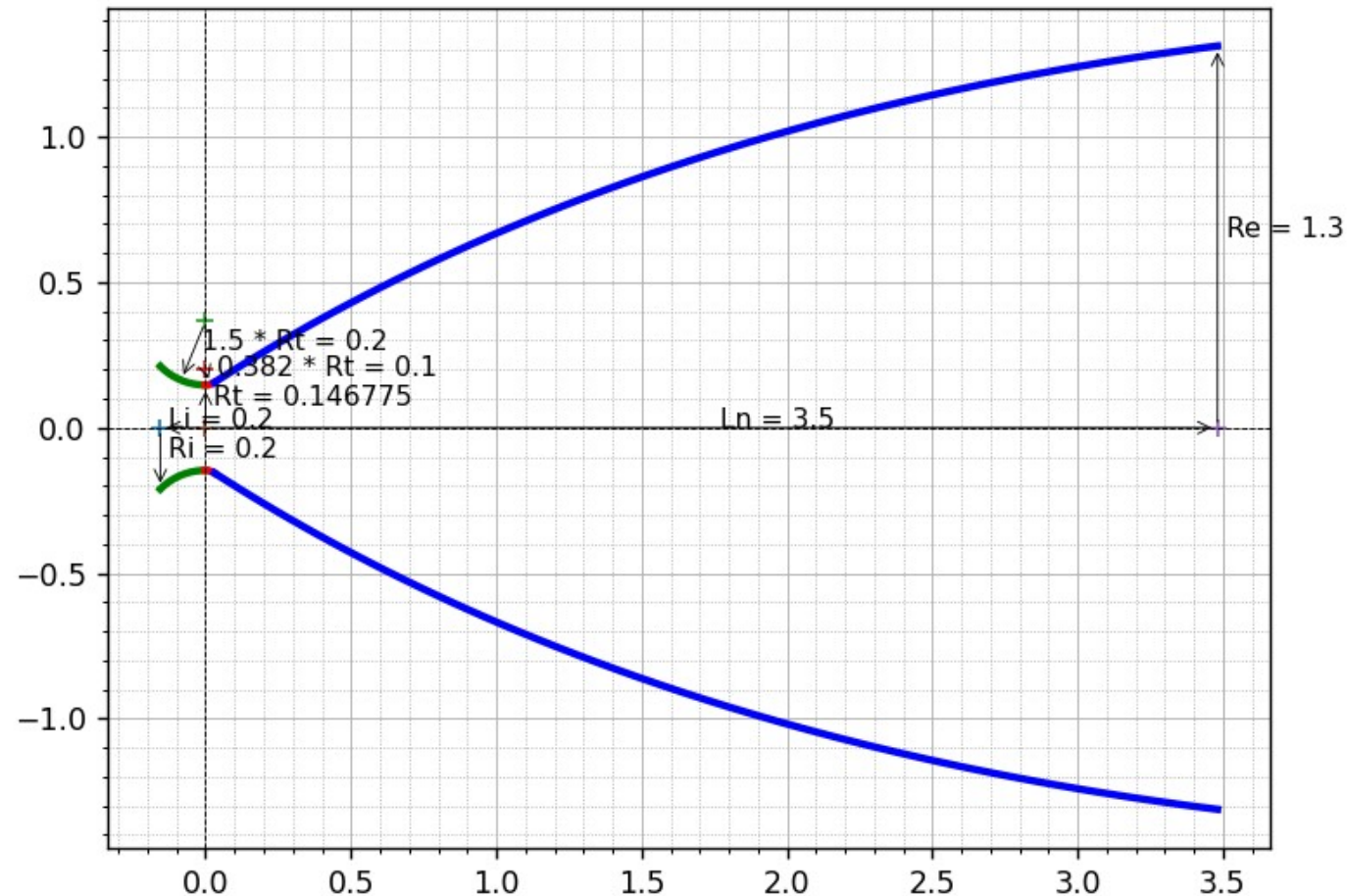


$$\theta_E = 0.1275835683207855, \theta_N = 0.5630432166933707$$

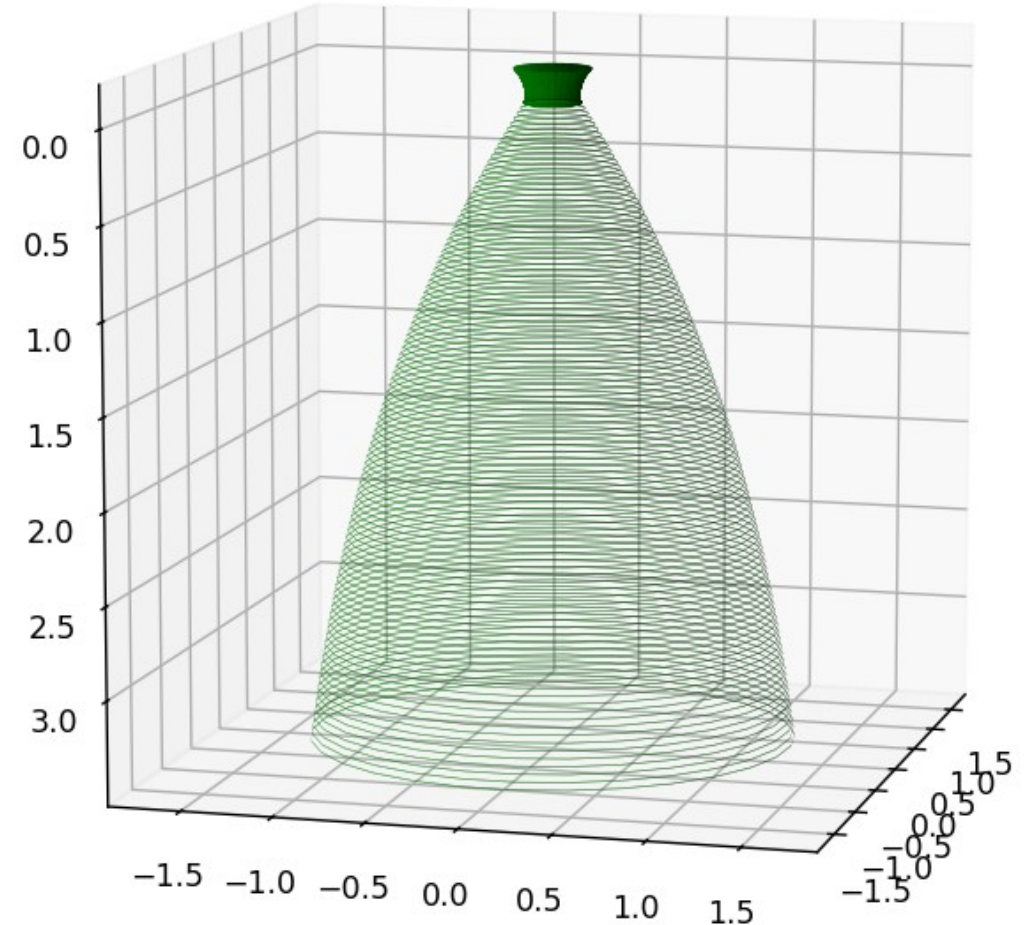


Nozzle Design: Area Ratio = 80

Bell Nozzle
[Area Ratio = 80, Throat Radius = 0.146775]



$$\theta_E = 0.12566370614359174, \theta_N = 0.5707226654021458$$



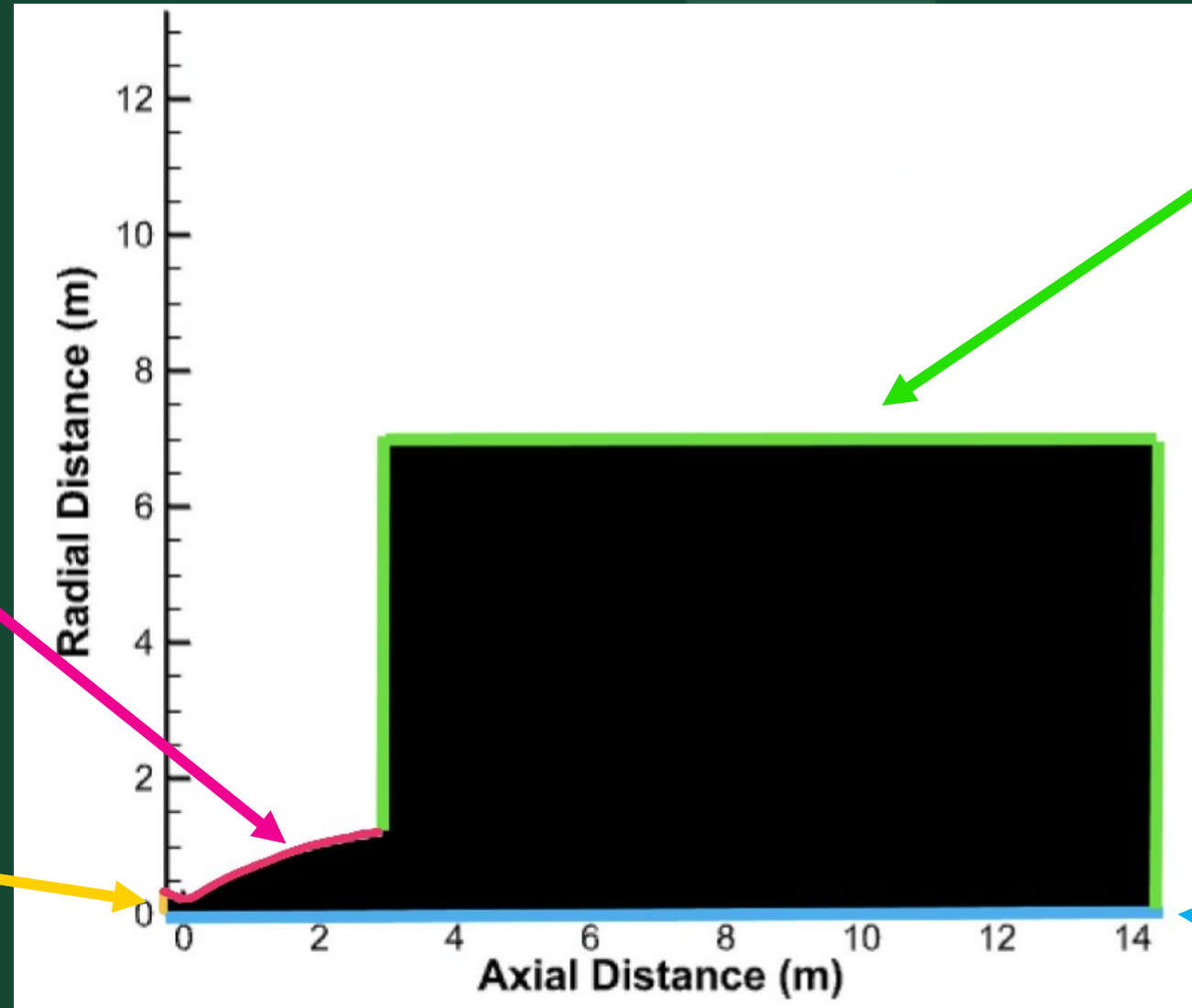
Boundary Conditions

Wall:
Stationary, No Slip

Inlet:
20,643,000 Pa
3589.15 K

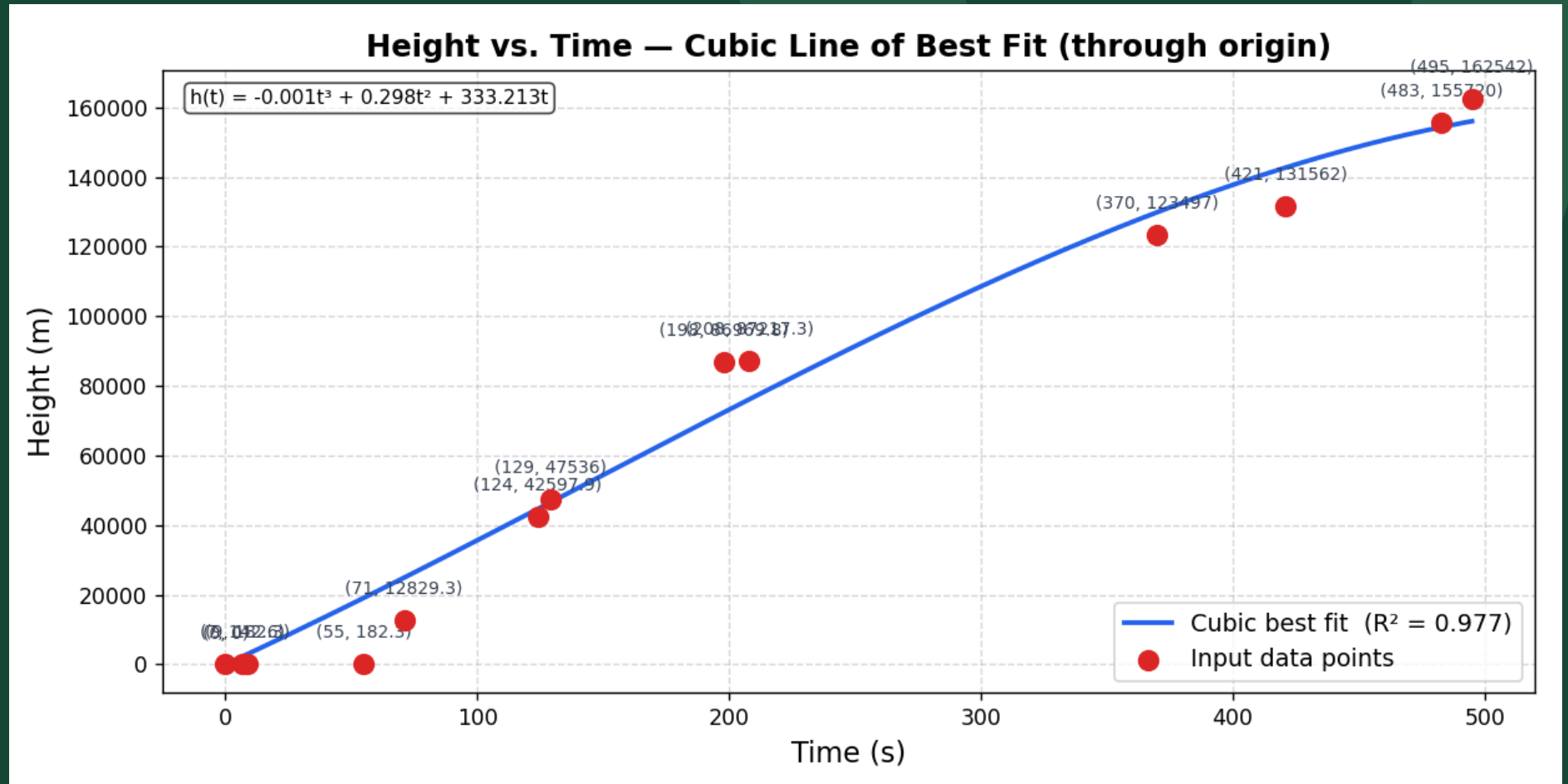
Outlet:
transient.c
300 K

Axisymmetric
c



Boundary Conditions

```
DATA_POINTS = [  
    (0.0, 0.0),  
    (7.0, 142.6),  
    (9.0, 182.3),  
    (55.0, 182.3),  
    (71.0, 12829.3),  
    (124.0, 42597.9),  
    (129.0, 47536.0),  
    (198.0, 86969.8),  
    (208.0, 87217.3),  
    (370.0, 123497.3),  
    (421.0, 131561.7),  
    (483.0, 155719.6),  
    (495.0, 162541.6),  
]
```



Boundary Conditions

For $h > 25000$ (Upper Stratosphere)

$$T = -131.21 + .00299 h$$

$$p = 2.488 * \left[\frac{T + 273.1}{216.6} \right]^{-11.388}$$

For $11000 < h < 25000$ (Lower Stratosphere)

$$T = -56.46$$

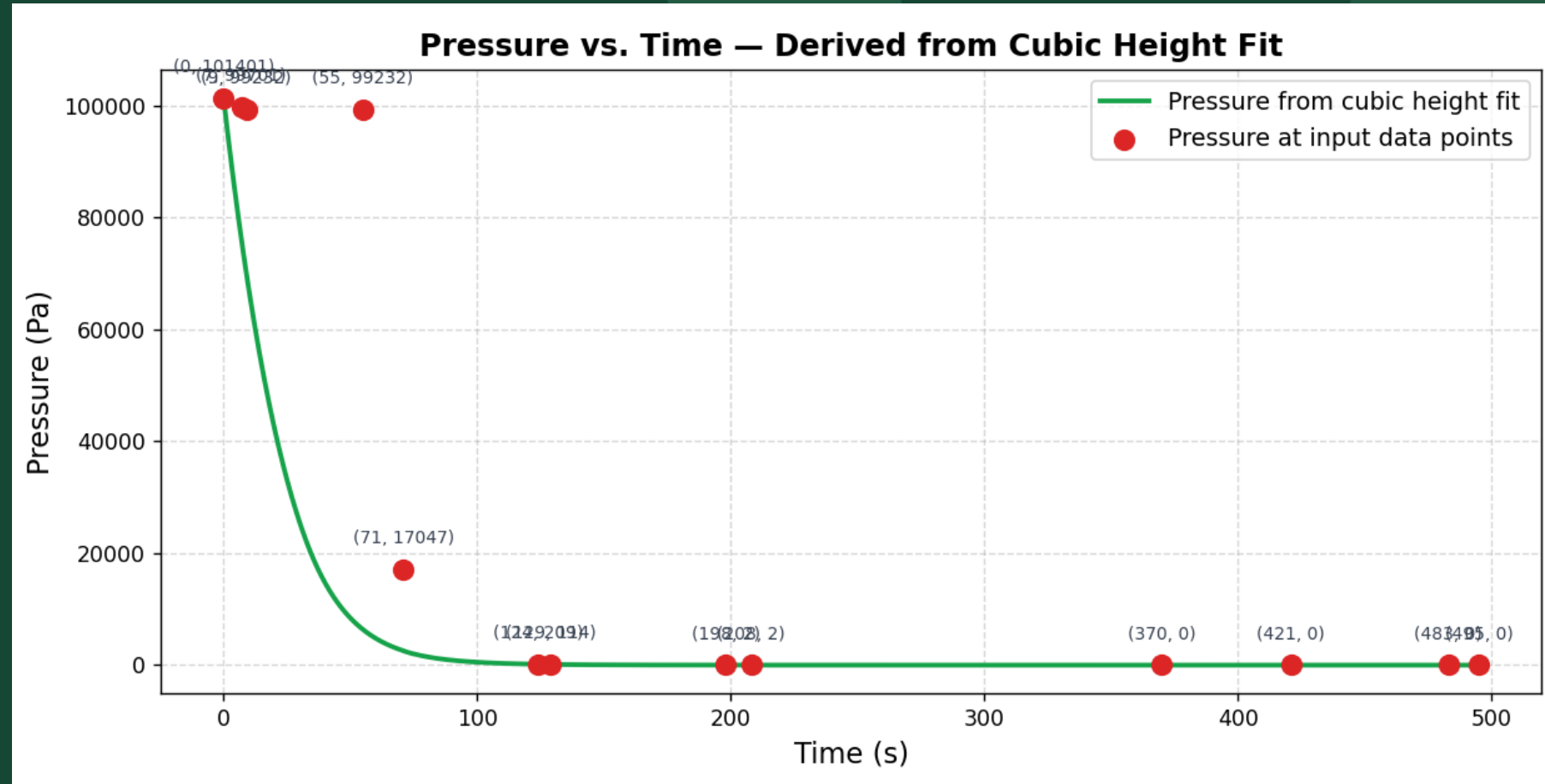
$$p = 22.65 * e^{(1.73 - .000157 h)}$$



For $h < 11000$ (Troposphere)

$$T = 15.04 - .00649 h$$

$$p = 101.29 * \left[\frac{T + 273.1}{288.08} \right]^{5.256}$$



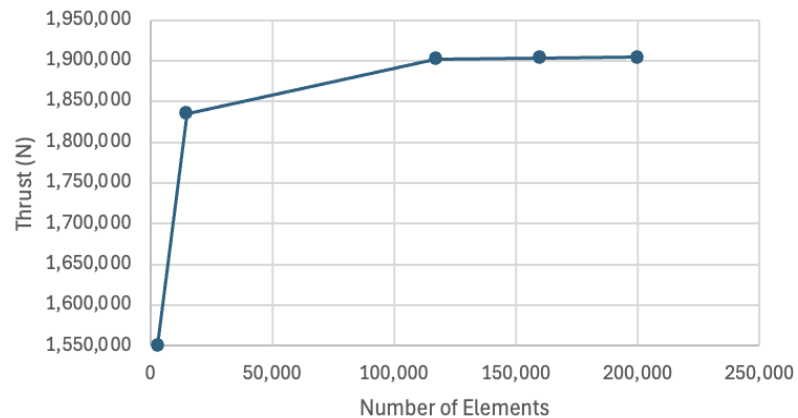
Mesh + Grid Sensitivity

Elements: 117,192
Thrust: 1,902,156.16

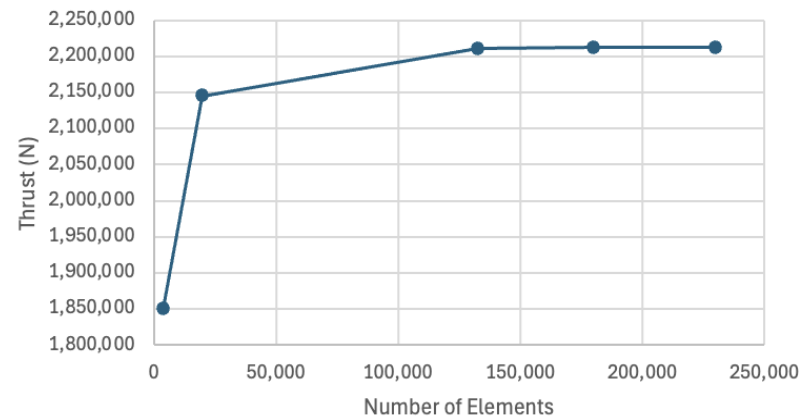
Elements: 132,822
Thrust: 2,210,515.77

Elements: 137,903
Thrust: 2,318,411

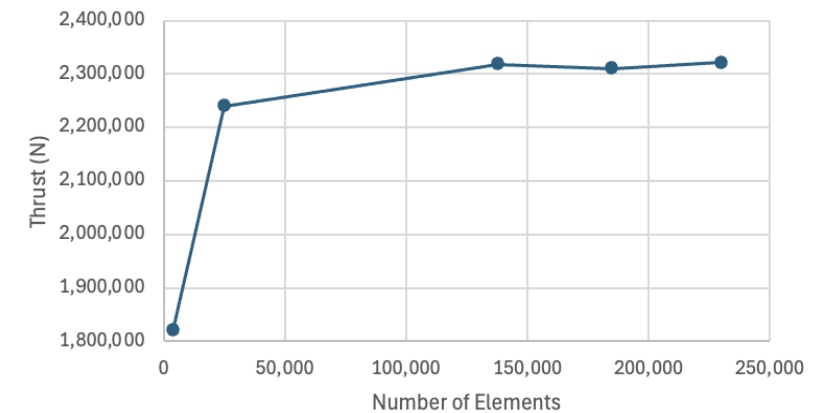
Mesh Convergence Study (Area Ratio 50)



Mesh Convergence Study (Area Ratio 69)



Mesh Convergence Study (Area Ratio 80)



NASA CEARUN

CEARUN Used to
Calculate all Propellant
Properties

Fuel: Liquid Hydrogen
Oxidizer: Liquid Oxygen
6:1 Oxidizer to Fuel Ratio

Initial Conditions for both
Fuel and Oxidizer Found
from Actual Artemis II
Data



THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 2994.0 PSIA

CASE = _____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2(L)	1.0000000	-9012.000	20.270
OXIDANT	O2(L)	1.0000000	-12979.000	90.170

O/F= 6.00000 %FUEL= 14.285714 R,EQ.RATIO= 1.322780 PHI,EQ.RATIO= 1.322780

	CHAMBER	THROAT	EXIT	EXIT
Pinf/P	1.0000	1.7403	1508.32	1172.29
P, BAR	206.43	118.61	0.13686	0.17609
T, K	3598.56	3381.52	1124.77	1184.91
RHO, KG/CU M	9.3928 0	5.7966 0	2.0651-2	2.5222-2
H, KJ/KG	-986.31	-2161.61	-10857.8	-10686.3
U, KJ/KG	-3184.04	-4207.89	-11520.5	-11384.5
G, KJ/KG	-62769.7	-60218.7	-30168.8	-31030.0
S, KJ/(KG)(K)	17.1689	17.1689	17.1689	17.1689
M, (1/n)	13.614	13.740	14.111	14.111
(dLV/dLP)t	-1.01898	-1.01413	-1.00000	-1.00000
(dLV/dLT)p	1.3293	1.2607	1.0000	1.0000
Cp, KJ/(KG)(K)	7.3163	6.6974	2.8281	2.8734
GAMMAS	1.1475	1.1487	1.2632	1.2580
SON VEL,M/SEC	1588.0	1533.2	915.0	937.2
MACH NUMBER	0.000	1.000	4.856	4.700

TRANSPORT PROPERTIES (GASES ONLY)

CONDUCTIVITY IN UNITS OF MILLIWATTS/(CM)(K)

VISC,MILLIPOISE 1.0832 1.0349 0.42252 0.44326

WITH EQUILIBRIUM REACTIONS

Cp, KJ/(KG)(K)	7.3163	6.6974	2.8281	2.8734
CONDUCTIVITY	14.2084	12.3504	1.7199	1.8208
PRANDTL NUMBER	0.5578	0.5612	0.6948	0.6995

Fluid: Hydrolox

Density:
Ideal Gas

Cp:
Piecewise Linear

**Thermal
Conductivity:**
Piecewise Linear

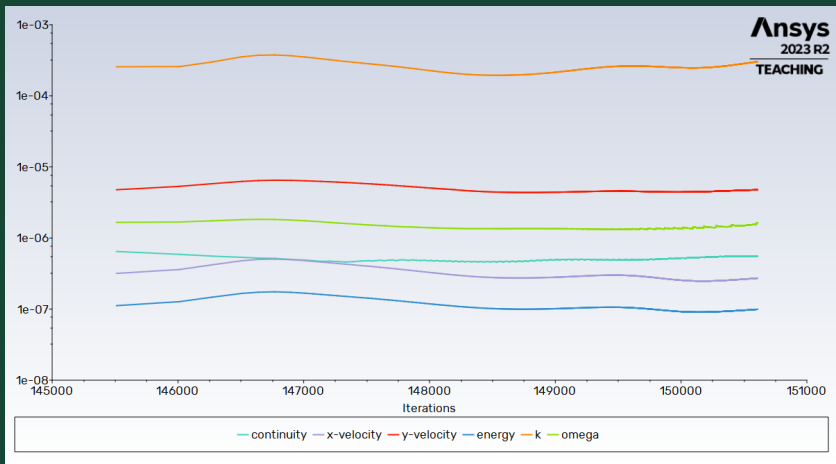
Viscosity:
Piecewise Linear

Molecular Weight:

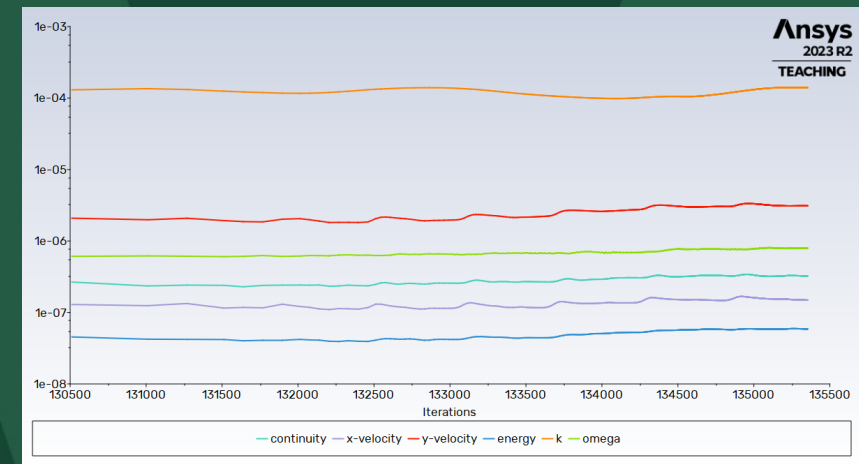
Area Ratio	T1	T2	Cp	Therma l Cond.	Viscosity
Pt 1: 50	900.3 2	3589.1 5	2987. 9	20.892	4.956e-5
Pt 1: 60	856.8 5	3589.1 5	2943. 1	19.816	4.751e-5
Pt 1: 69	825.0 3	3589.1 5	2909	19.021	4.595e-5
Pt 1: 80	792.7 1	3589.1 5	2873. 4	18.21	4.433e-5

Convergence

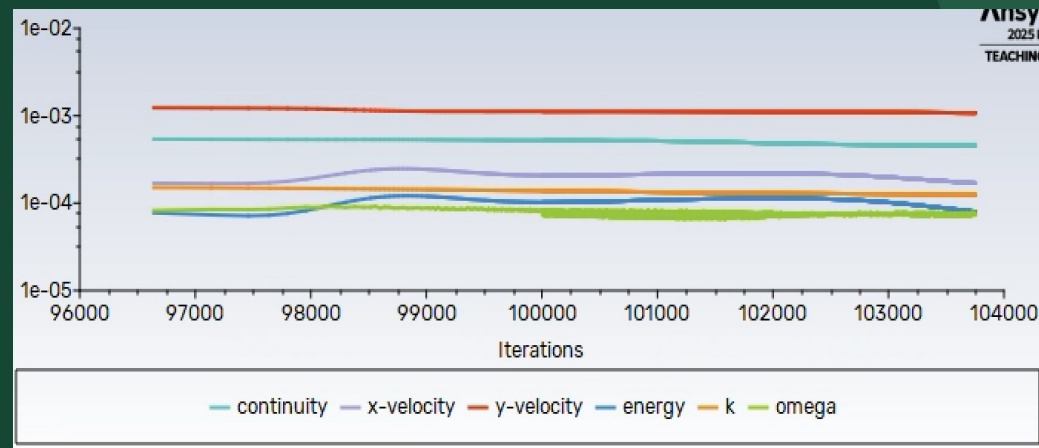
Area Ratio = 50



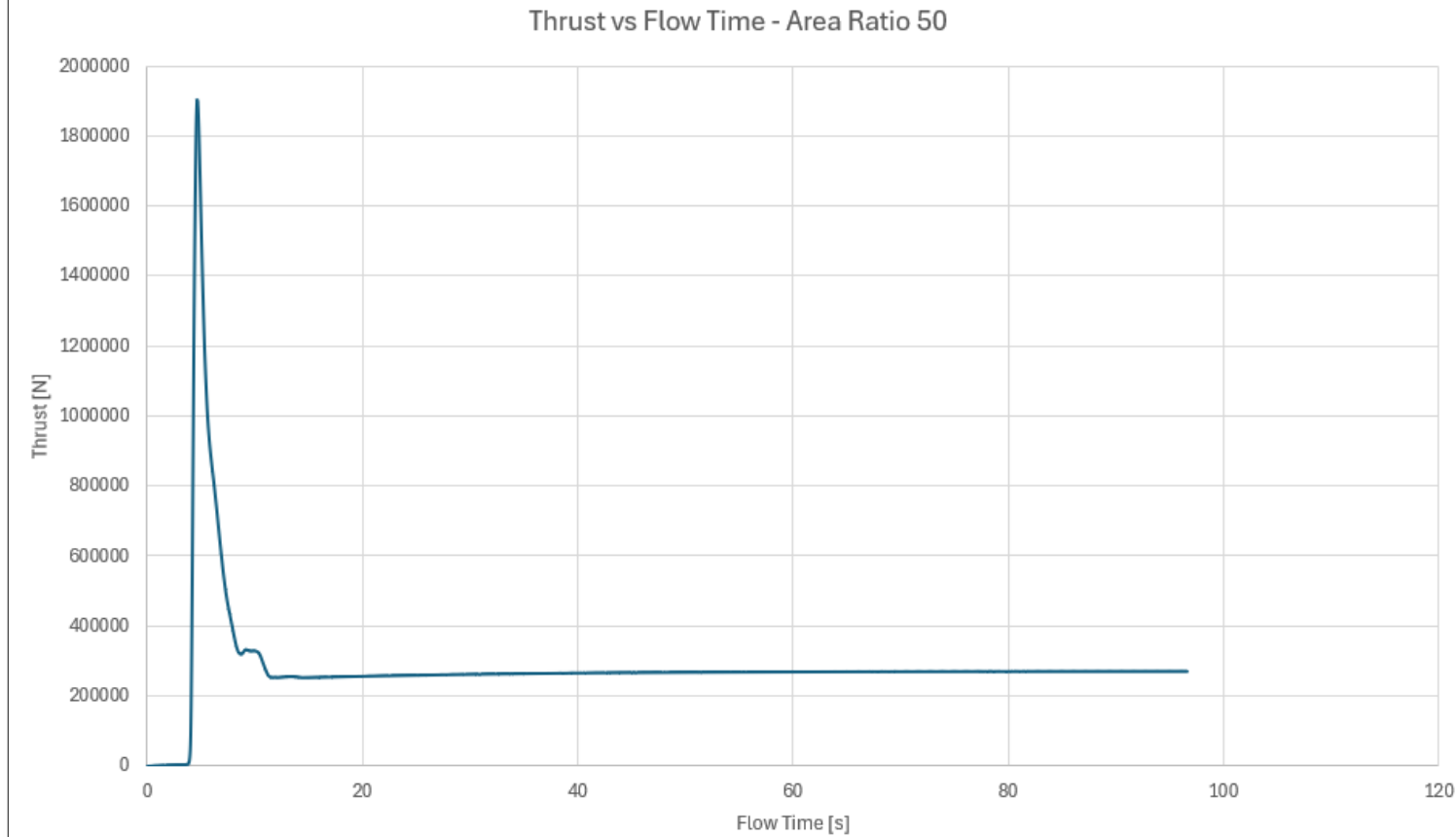
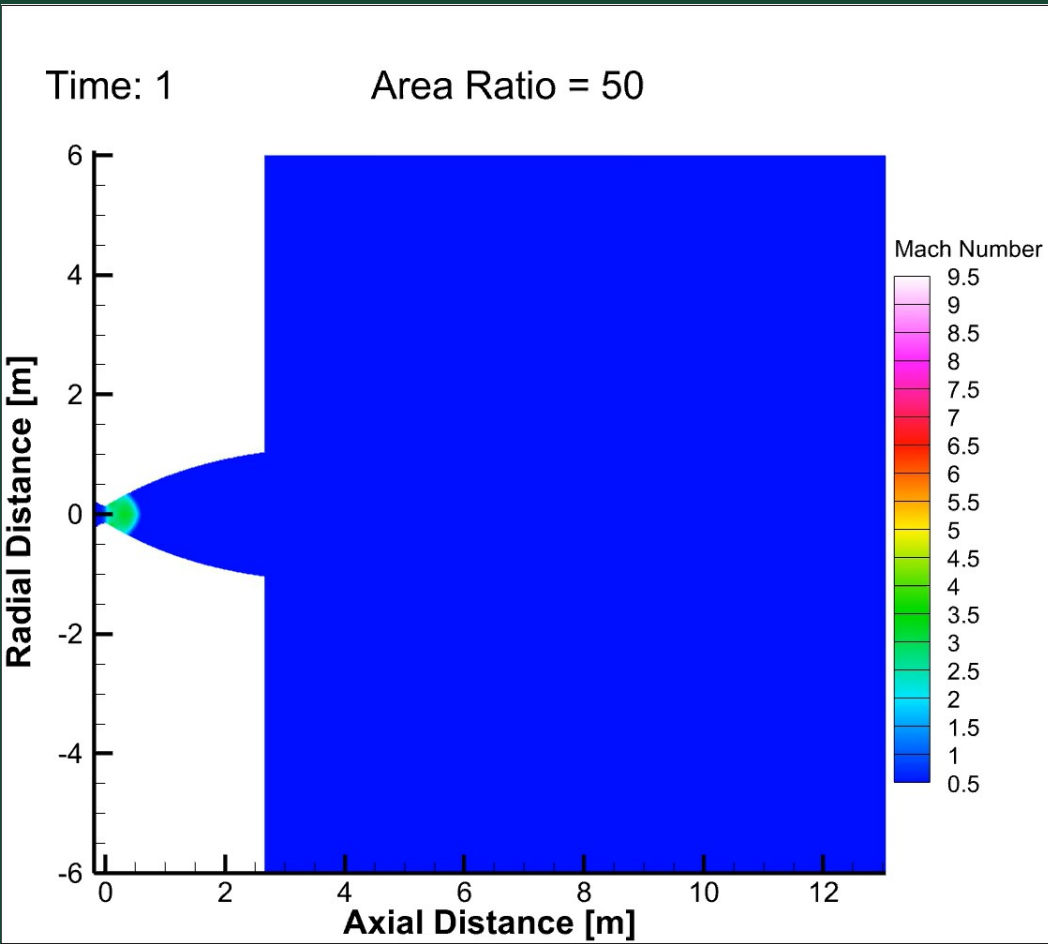
Area Ratio = 69



Area Ratio = 80

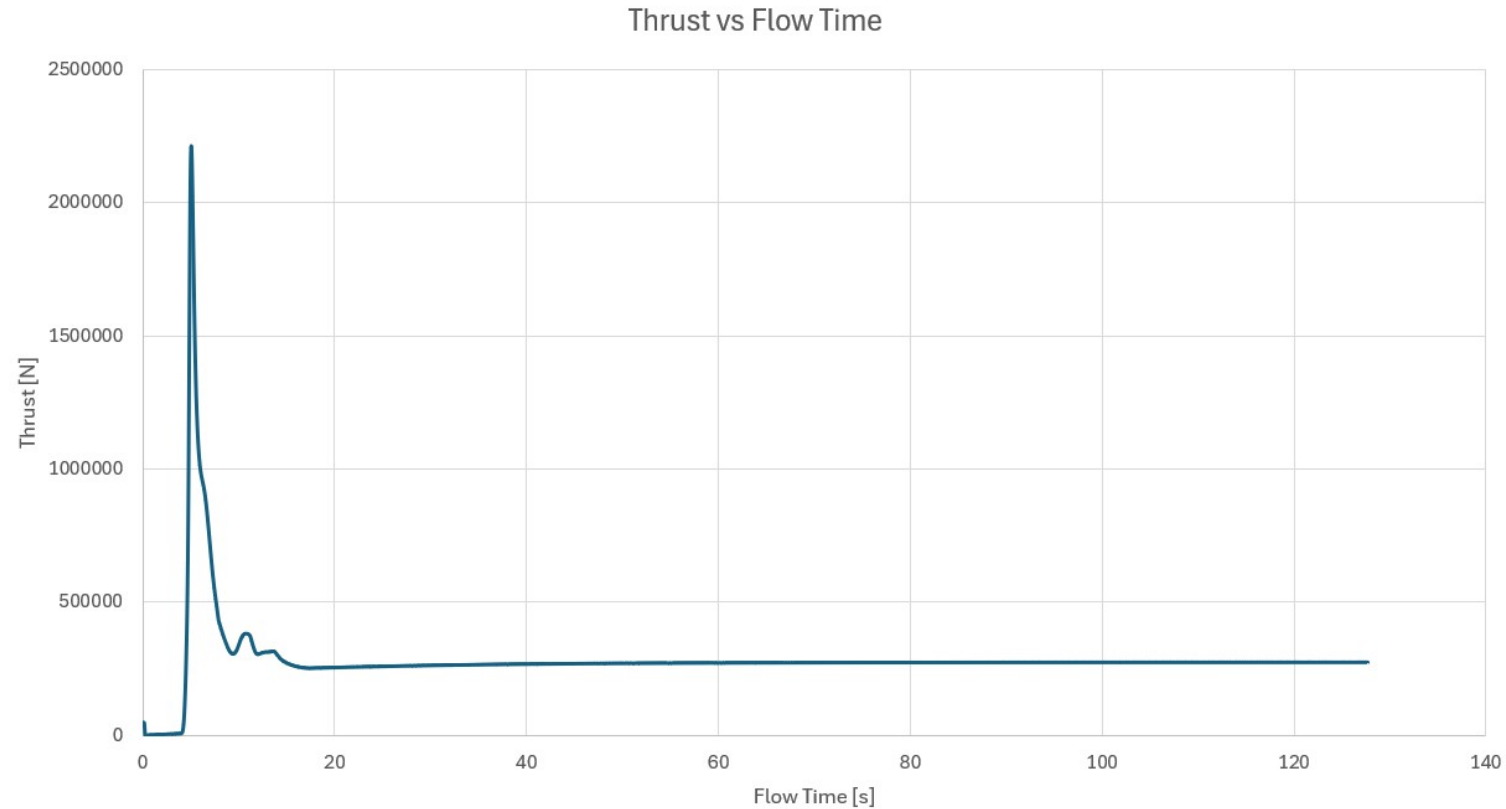
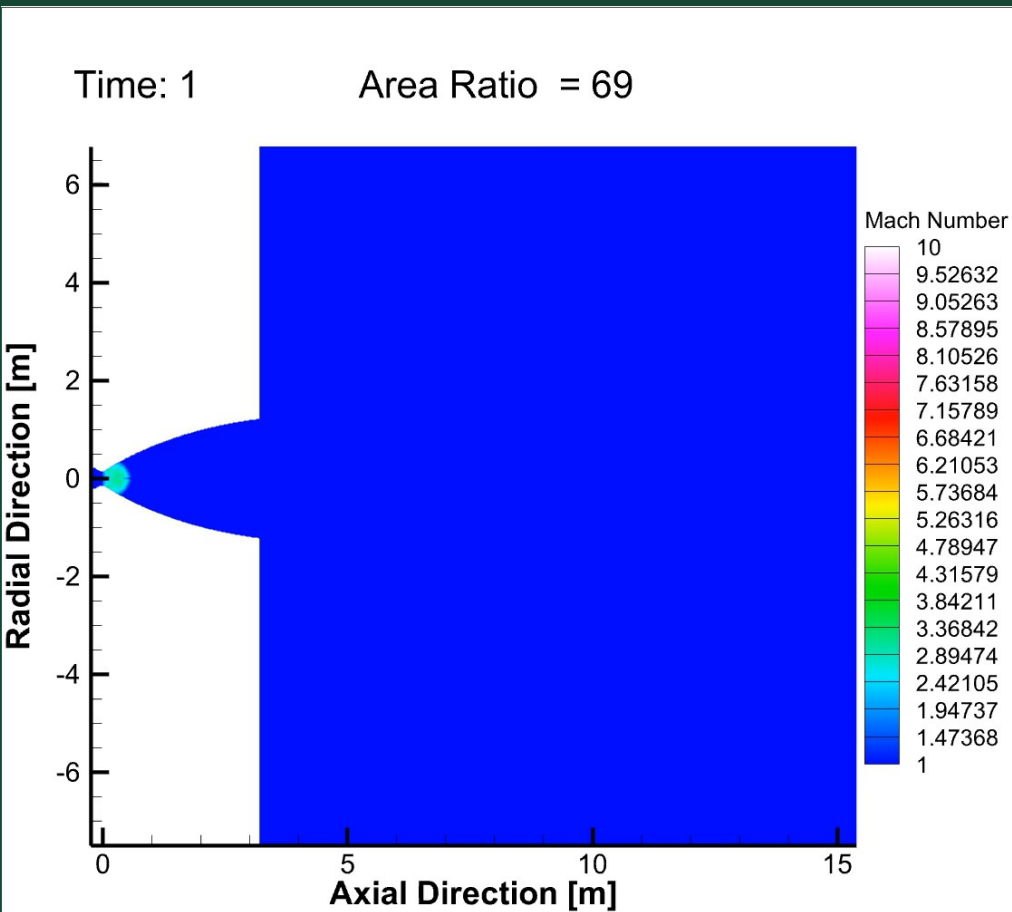


Results: Area Ratio = 50



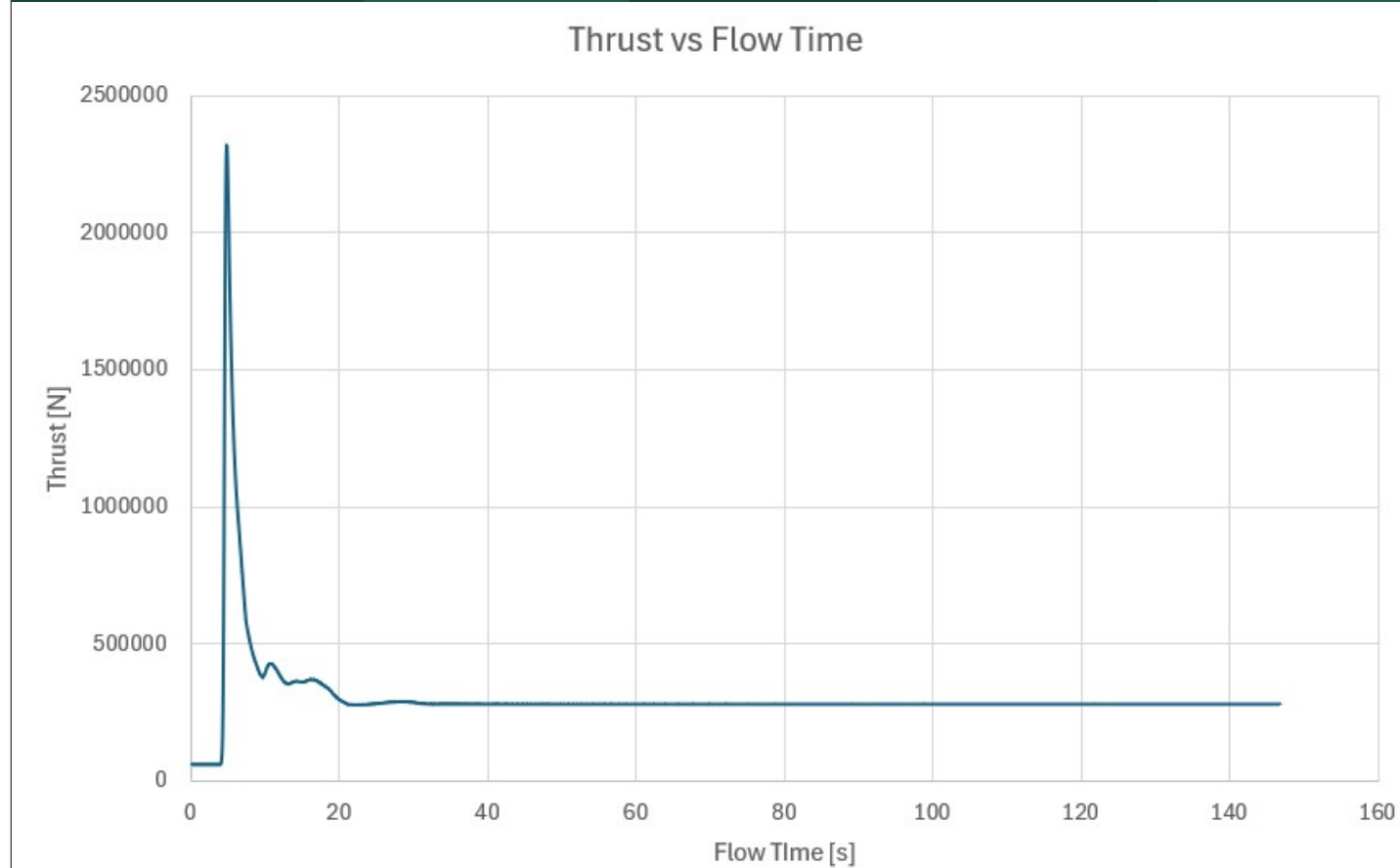
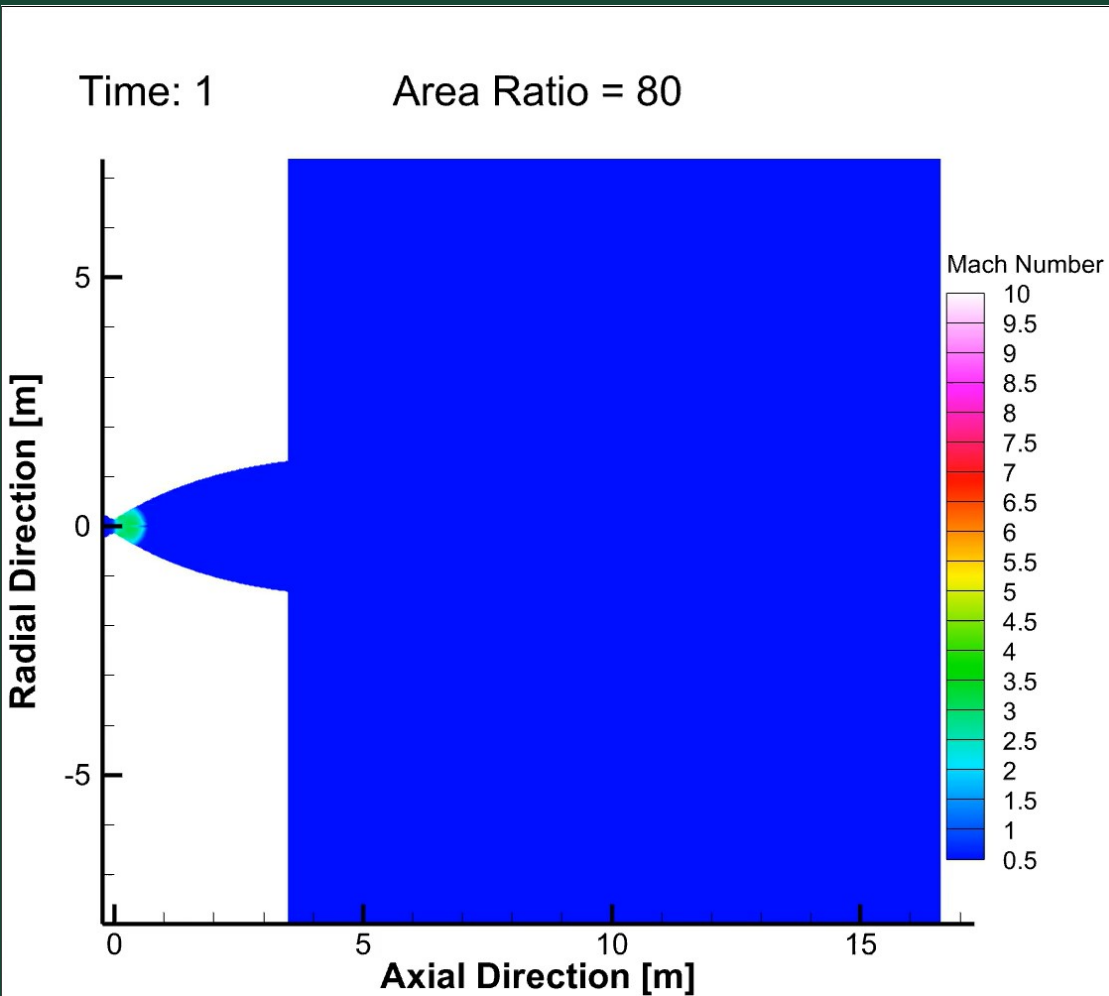
Max Thrust: 1902156.15672262 N at
4.65 s

Results: Area Ratio = 69



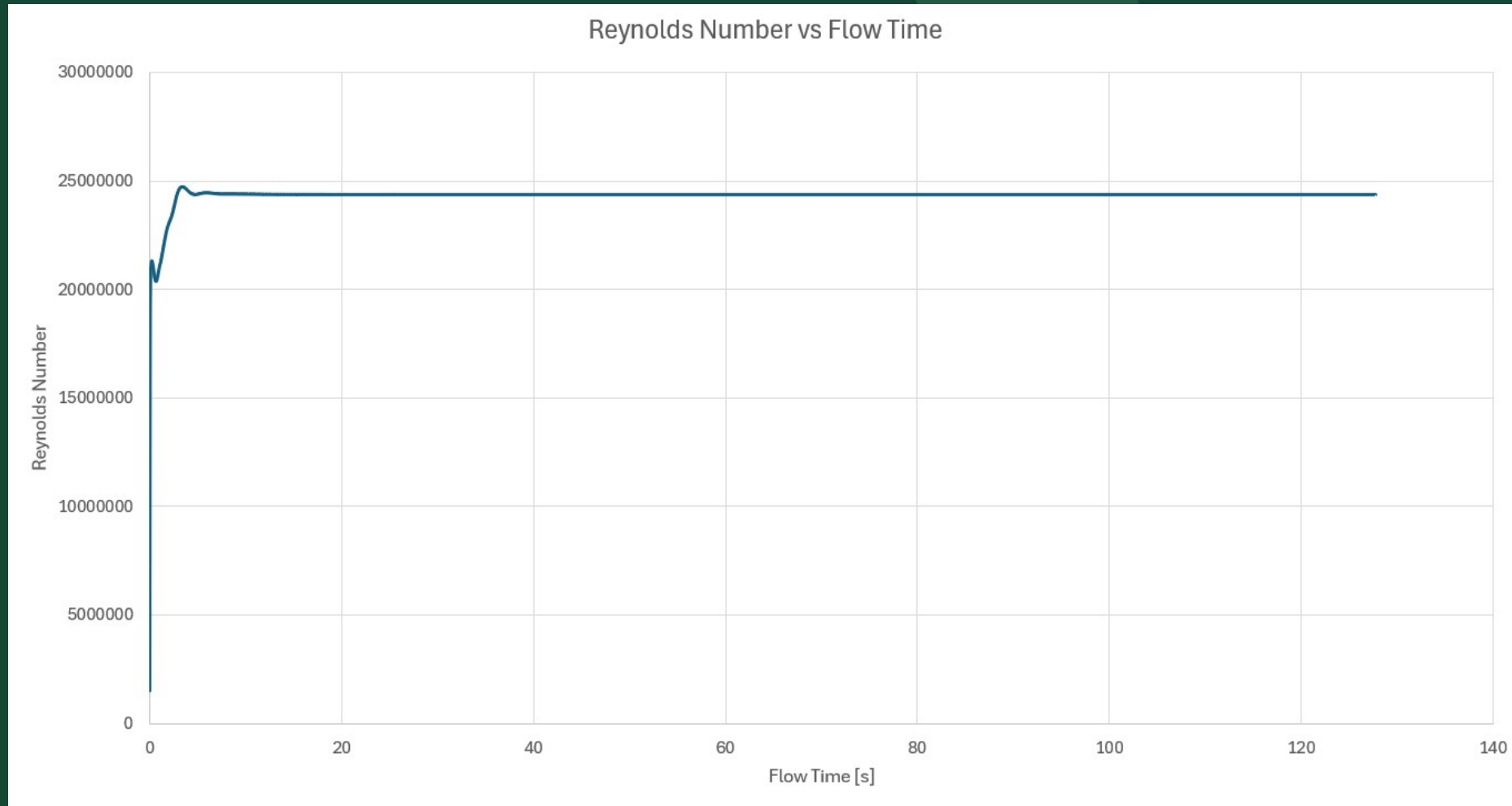
Max Thrust: 2210515.768 N at 5 s

Results: Area Ratio = 80



Max Thrust: 2318411.24136742 N at
4.8 s

Reynolds Number at Throat



Real vs Simulation Data

AREA RATIO	MAX THRUST (ATM) [N]	MAX THRUST (ATM) [LBF]	THRUST (VAC) [N]	THRUST (VAC) [LBF]	ACTUAL MAX (ATM) [LBF]	ACTUAL MAX (VAC) [LBF]
50	1,902,156.16	427,621.72	267,524.42	60,141.88	—	—
69	2,210,515.77	496,943.72	274,094.50	61,618.89	418,000	512,000
80	2,318,411.24	521,199.58	276,237.37	62,100.63	—	—

Conclusion

- Maximum thrust for each nozzle occurred in troposphere range
- The maximum thrusts determined were near expected values while in the atmosphere, however produced much less thrust in a vacuum.
- To further improve this study, a much longer simulation is required. Increasing the number of iterations would refine flow geometries that were unrealistic.

Work Cited

- [1] National Aeronautics and Space Administration, "Earth's Atmosphere," NASA Glenn Research Center. [Online]. Available: <https://www.grc.nasa.gov/www/k-12/airplane/atmosmet.html>. [Accessed: May 3, 2026].
- [2] R. Ram, "Bell-Nozzle," GitHub repository. [Online]. Available: [Bell-Nozzle GitHub Repository](#). [Accessed: May 3, 2026].
- [3] R. Nakka, "Thrust Optimised Parabolic Nozzle," AspireSpace. [Online]. Available: Thrust Optimised Parabolic Nozzle PDF. [Accessed: May 3, 2026].
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- [5] L3Harris Technologies, "RS-25 Rocket Engine Specification Sheet," July 2024. [Online]. Available: <https://www.l3harris.com/sites/default/files/2024-07/l3harris-ar-rs-25-spec-sheet.pdf>. [Accessed: May 1, 2026].
- [6] L3Harris Technologies, "RS-25 Engine." [Online]. Available: <https://www.l3harris.com/all-capabilities/rs-25-engine>. [Accessed: May 3, 2026].
- [7] National Aeronautics and Space Administration, "Space Launch System: RS-25 Core Stage Engine Fact Sheet," July 31, 2015. [Online]. Available: https://www.nasa.gov/wp-content/uploads/2020/09/final_sls_rs25_fact_sheet_508_07312015.pdf. [Accessed: May 4, 2026].
- [8] National Aeronautics and Space Administration, "Chemical Equilibrium with Applications (CEA)," NASA Glenn Research Center. [Online]. Available: <https://cearun.grc.nasa.gov/>. [Accessed: May 6, 2026].